

Arnav Karekar

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EDUCATION

Texas A&M University, GPA: 3.53 – College Station, TX

Bachelor of Science in Computer Science – August 2024 – May 2028

TECHNICAL SKILLS

Certifications: CompTIA Security+, ISC2 Certified in Cybersecurity, AWS Certified Developer – Associate

Languages: Python, Java, JavaScript, C++, SQL, PowerShell, HTML/CSS

Developer & Security Tools: Microsoft Defender, Microsoft Intune, Proofpoint Endpoint DLP, Elastic Security, Cisco Identity Intelligence, Splunk, Security Onion, Palo Alto Firewall

EXPERIENCE

Undergraduate Researcher | Concept Drift, Malware Detection, Android Security – May 2025 – Present

Department of Computer Science & Engineering at Texas A&M – College Station, TX

- Improved malware detection reliability under data drift by building a Python pipeline that extracted features from DREBIN and AndroZoo and modeled temporal app arrival patterns across simulated deployment windows.
- Identified the most resilient retraining strategy for production-style **distributed systems** handling streaming data by designing custom concept-drift benchmarks that compared accuracy decay across ADWIN, TRANSCENDENT, and aggressive retraining.

Student Technician | Proofpoint, Elastic Security, Cisco Identity Intelligence – October 2025 – Present

Department of Technology Services at Texas A&M – College Station, TX

- Training for: **Cisco Certified Network Associate (CCNA)**.
- Reduced exposure risk from sensitive data entering generative AI applications, as measured by 6 potential-exposure reports generated and escalated to the DLP team, by performing frontline triage and analysis in Proofpoint Endpoint DLP.
- Supported enterprise **vulnerability management** and identity risk reduction by investigating anomalous access patterns in Elastic Security SIEM and correlating findings with Cisco Identity Intelligence.

Cybersecurity Intern | Microsoft Defender, Microsoft Intune, PowerShell – June 2025 – August 2025

Arkansas Department of Education – Little Rock, AR

- Raised the organization's Microsoft Defender Secure Score by 6 points, as measured by pre/post Secure Score benchmarking, by developing and deploying automated PowerShell remediation scripts across the environment.
- Built hands-on **incident response** capability against live network and application-layer attacks, as measured by successful containment of 3 simulated attack scenarios, by completing live-fire exercises in the Cloud Range platform using Splunk, Security Onion, and Palo Alto Firewall—deepening familiarity with the current **threat landscape** facing K-12 and public-sector networks.

PROJECTS

PhishShield Chrome Extension | JavaScript, React, Python, Cloudflare Workers – March – May 2026

- Developed a Manifest V3 Chrome extension that detects malicious links in real time using a four-layer pipeline of 14 URL heuristics, an 18-feature in-browser neural network, DOM inspection, and download interception.
- Integrated Google Safe Browsing, PhishTank, and domain-age intelligence through a Cloudflare Worker and built a React interface that displays risk scores, explains detected threats, and syncs user whitelists and analysis settings.

RocketSploit Website | Python, Flask, SQL, Git – February – March 2024

- Built a complex website featuring numerous web hacking challenges such as cookie manipulation, directory traversal, and cross-site scripting.
- Helped high school competitors gain a better understanding of hacking and vulnerabilities through various web hacking and data sniffing challenges.

Personal Network Attached Storage | TrueNAS, NAS, Troubleshooting – November 2024 – Present

- Built and managed a TrueNAS-based media server to stream TV shows and movies to multiple users.
- Set up and maintained an image backup system for reliable local data redundancy.